USADEBUSK

Heater Pigging & Decoking Services

Valero Texas City Refinery

Mechanical pigging & decoking for fired heaters

Presented by Jason Harman
Commercial Manager
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Agenda

Heater pigging & decoking at Texas City

USA DEBUSK

01 Pigging & decoking methodology

02 Execution

03 Readiness & resources

04 Equipment & technology

05 Cleanliness verification & QA/QC

06 Emergency response

07 Safety & risk mitigation

08 Capabilities

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10 Why USA DeBusk

SERVICES

Our services for Texas City

Mechanical decoking of fired heater coils, planned or on short notice

Coker heaters

Hard coke in radiant and convection coils, on run-length or turnaround cycles

Crude & vacuum heaters

Pitch, resid and coke across mixed-diameter passes

Charge & process heaters

Process heater coils cleaned and verified ahead of restart or inspection

Smart pig support

Coils cleaned to inspection standard; spare TriMax pump drives the inspection tool

Planned

turnarounds & run-length decokes

Short notice

emergency mobilization

Optional

closed-loop filtration



01 METHODOLOGY

Bi-directional mechanical decoking

Every pass follows the same controlled sequence

1. **Rig-in & pressure test**
   Connect 300# launchers and receivers; fill and test before any launch
2. **Baseline flow test**
   Record pressure at a set rate for a before-and-after comparison
3. **Gauge foam run**
   Proves the pass is open and sets the initial pig size
4. **Scraper pig progression**
   Carbide-studded pigs run in both directions, stepped up ⅛" per clean pass
5. **Oversize polish**
   Up to tube ID + ¼" maximum, with real-time pressure monitoring
6. **Gauge foam & sign-off**
   Line-size foam witnessed by Valero, final flow test, pass signed clean



Pig progression plan

The gauge foam run sets the starting size; the smallest tube ID sets the ceiling

Example: 4" Sch 80 coil

Tube ID 3.826"



Example: mixed-diameter pass, 3" to 8"

Smallest ID 2.900" governs the whole pass



The sizing rule

Max pig OD = smallest tube ID + 0.250"

- Step up ⅛" after each clean pass
- Pig wear drives size & appendages
- Body length set for bends & mule ears
- Gauge foam sets the starting size
- Line-size foam witnessed at sign-off

01 METHODOLOGY

Engineered decoking pigs, matched to the coke

The most diverse selection of engineered decoking pigs in the world



New engineered decoking pigs

Soft-core and hard-core designs

Urethane or high-density foam bodies, plus foam pushers for flushing and proving the pass

A full range of durometers

Soft to hard compounds, chosen to compress through tight bends or cut hard coke

Bald or studded

Tungsten carbide, hardened-steel and stainless stud options, matched to the tube metallurgy

Heavy-coke designs

High-durometer, fully studded pigs brought in when pig wear shows hard, heavy coke

01 METHODOLOGY

Coil-specific controls

Every heater is planned from its drawings before we mobilize

Tube geometry

Footage, tube IDs and diameter changes per pass set the pig progression and the duration

Metallurgy

Stainless and chrome coils: water quality verified before fill, no galvanized contact with tubes

Connections

Coil flange size and rating confirmed; spools and adapters matched to our 300# launchers and receivers

Return bends

Pig length and flexibility selected for 180° returns, plug headers and mule ears

Launcher access

Inlet and outlet elevations walked down; rig layout set before rig-in

Coke & service

Coker, crude or vacuum service sets the pig set staged on day one

02 EXECUTION

How a decoke runs

Continuous 24/7 pigging on 12-hour shifts; passes pigged in conjunction

Plan

Drawings reviewed, pig progression and rig layout set

Rig-in

Launchers and receivers connected, filled and pressure tested

Pig

Up to three passes at once per TriMax, around the clock

Rig-over

Move to the next set of passes, if the heater has more

Verify

Final gauge foam and flow test, pass signed off with Valero

Rig-out

Equipment out, job book delivered

24/7

two 12-hour shifts until the heater is handed back

Up to 3

passes pigged at once on each TriMax

2+

TriMax units on larger heaters or multi-heater jobs



03 READINESS & RESOURCES

Crew & equipment, every job

Equipment

1	TriMax triple pumper
1	Support unit: fittings, tooling, spares
2	Crew trucks

Manpower

2	Supervisors: 1 day, 1 night
3	Operators: 2 day, 1 night
1	Project manager

~5%
crew turnover, last 12 months

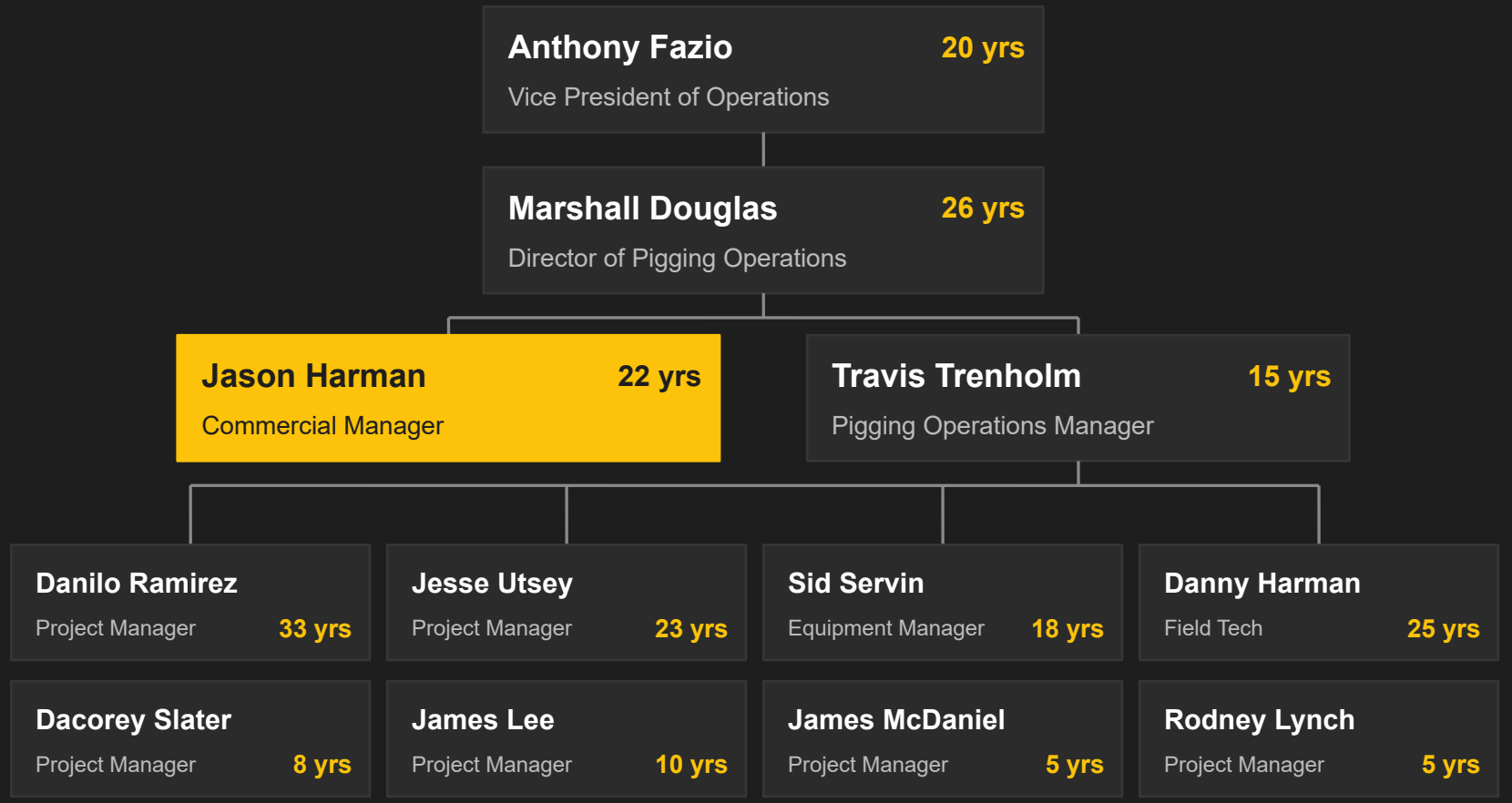
24/7
two 12-hour shifts

100%
training verified before mobilization

Our pigging team

210
years of combined pigging & decoking experience

15+
years self-performing heater pigging



The TriMax triple pumper

6 identical TriMax units, 18 engines: one standardized fleet

Three independent pump systems

Closed-loop, bi-directional; 50% more productive than a twin pumper

Automated HMI control

Three 17" touchscreens, automated digital pig logs

150–300 psi working pressure

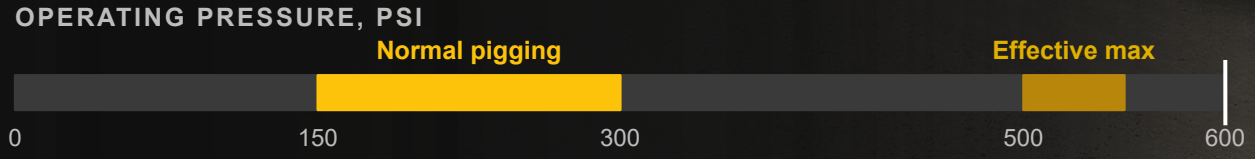
Pump rated 600 psi; 500–550 psi effective after valve losses

Real-time sensor data

Coke shows as pressure spikes at the operator interface

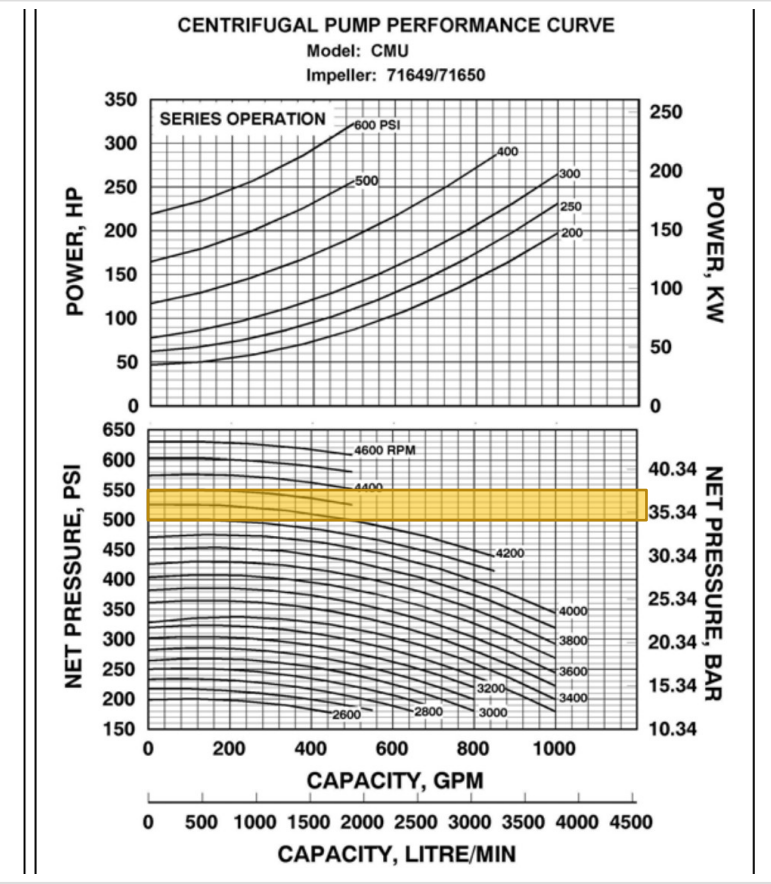
Cummins QSL9, 333 bhp, Tier 4 Final

The same engine across all 18 pumps in the fleet



TriMax pump performance

Waterous CMU two-stage pump, one per pass: series-operation curve, as used for pigging



Manufacturer curve: Waterous Company, form F-2692

500–550 psi
Effective maximum at the pig, after losses through our valves

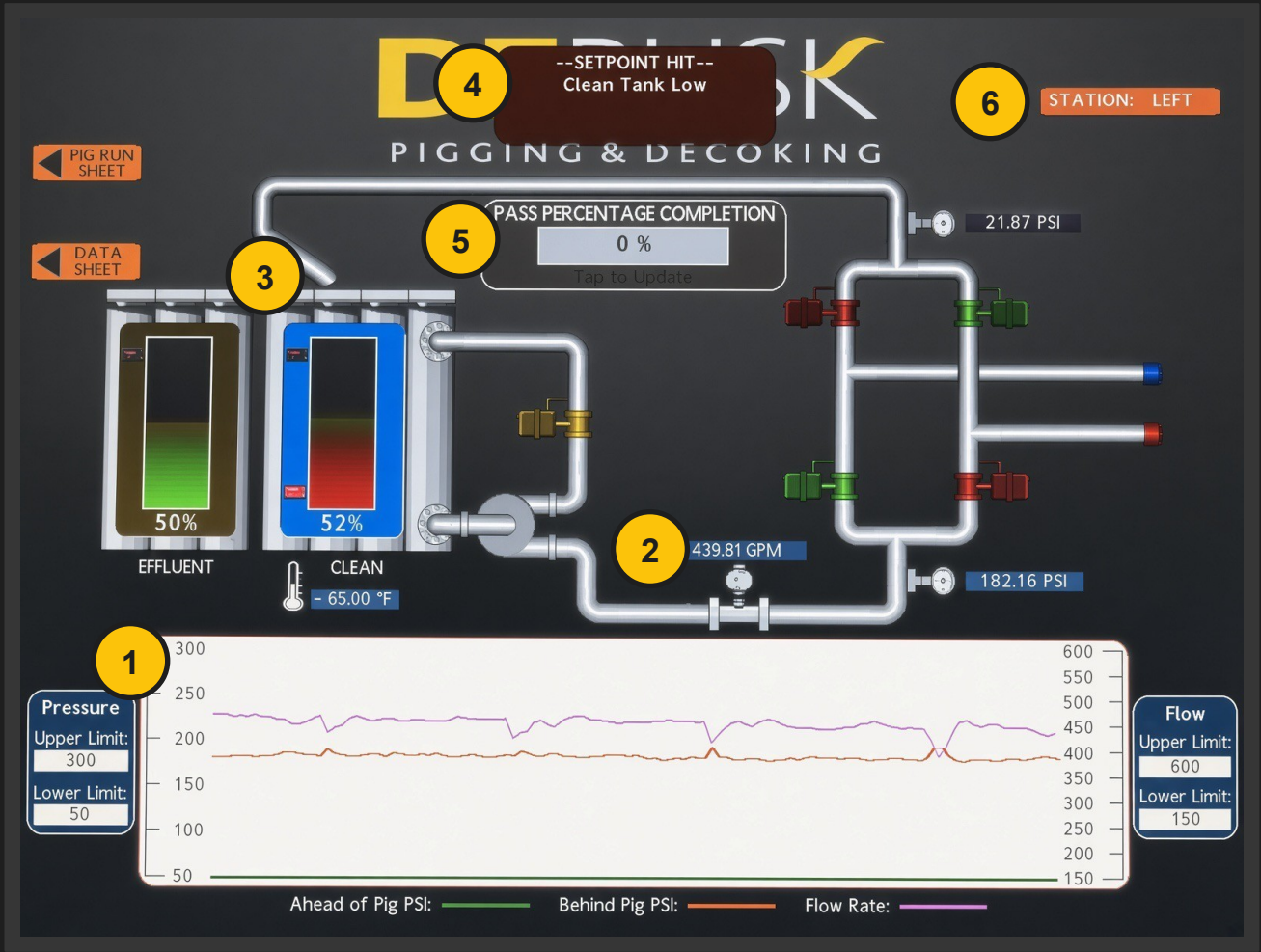
Reading the curve Each line is one engine speed; pressure falls as flow rises. Pigging runs well inside this envelope at 150 to 300 psi.

150–300
psi normal pigging

500–550
psi effective max, after valve losses

600
psi rated pump maximum

Real-time control at every operator station



1. **Live trends**
   Pressure ahead of and behind the pig, plus flow; coke shows as a spike
2. **Flow & pressure**
   GPM and psi at the pump, updated continuously
3. **Tank levels**
   Clean and effluent tanks on guided-radar indicators
4. **Setpoint alarms**
   Upper and lower limits on pressure and flow trigger alerts
5. **Digital pig logs**
   Pass completion, pig run sheet and data sheet recorded automatically
6. **One station per pass**
   Left, center and right stations run independently

Optional closed-loop filtration

Runs alongside pigging, independent of coil pressure



Filtrate returns to the clean tank; pigging continues without interruption

400 GPM

press capacity

1,243 ft²

filtration area per press

3

trailer-mounted presses in the fleet

Optional: priced separately

- **Reduces fresh-water demand**
- **Reduces wastewater volume**
- **Coke fines captured as filter cake for disposal**

05 CLEANLINESS VERIFICATION & QA/QC

Verifying every pass is clean

Four independent indicators, reviewed side by side with Valero

1 Sensor data

Coke shows as a pressure spike; the tech stops and scrubs that exact spot

2 Effluent returns

"Black" in the return stream fades to nothing as the pass cleans

Joint sign-off per pass



3 Pig condition

Wear marks guide pig size and appendage choices

4 Final gauge foam

Soft line-size (or slightly over) foam, witnessed by the Valero rep

Field-proven process: successful sign-offs from all of our clients and reliable first-attempt smart pig data.

05 CLEANLINESS VERIFICATION & QA/QC

QA/QC from walkdown to job book

Governed by the USA DeBusk QA/QC manual

Before

- Pre-job walkdown with operations
- Coil data checked against Valero records
- JSA, permits and isolation confirmed
- Pig progression staged by heater

During

- Daily report to one point of contact
- Automated digital pig log per pass
- Flow tests before and after at the same rate
- JSA reviewed at every shift change

Close-out

- Per-pass cleanliness sign-off
- Final pig size recorded per pass
- Job book delivered to Valero
- Post-job review and lessons learned

06 EMERGENCY RESPONSE

Based in Deer Park, TX

Responsive support for Texas City

Our fleet stages at our Deer Park base, a short drive from Texas City



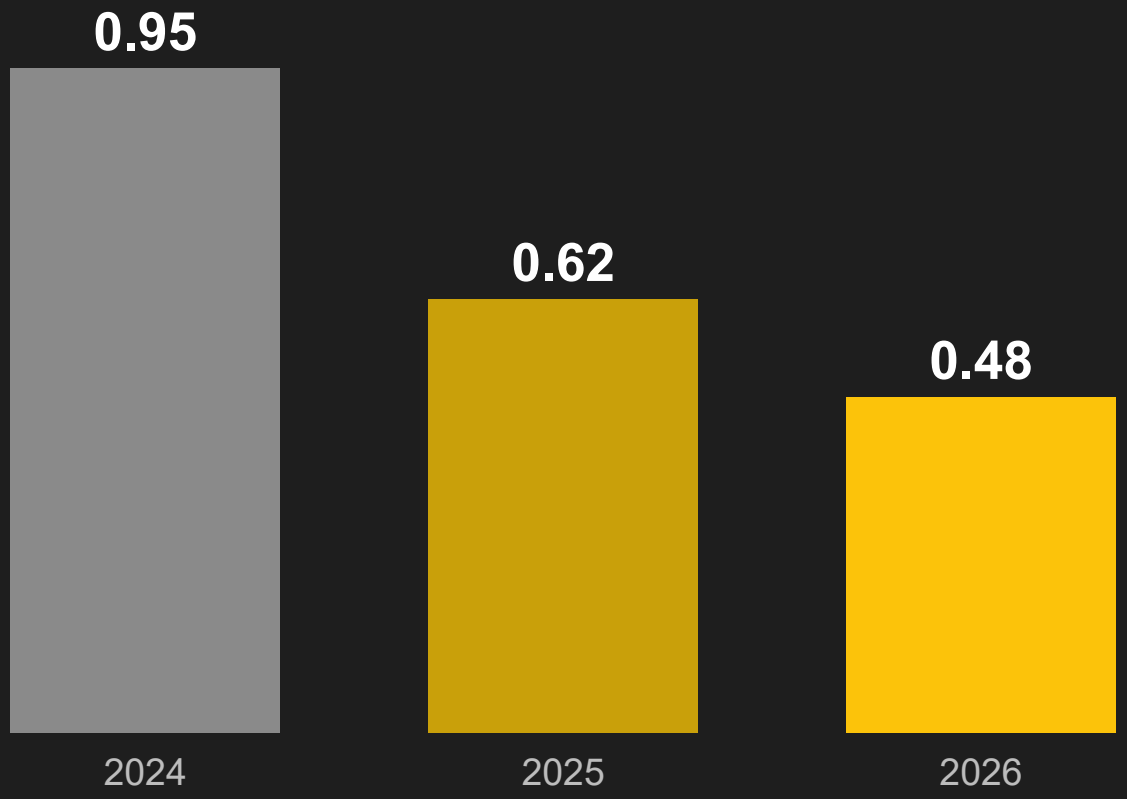
24/7
Dispatch and account manager, any hour

Same day
Response from Deer Park to Texas City

6
TriMax units staged at Deer Park between jobs

Safety performance

Experience Modification Rate: 1.0 is the industry baseline; lower is better



A
ISNetworld safety grade

0.48
EMR, 2026

−49%
EMR since 2024

0
fatalities in 3 years



07 SAFETY & RISK MITIGATION

Risk mitigation in the field

JSA & permits

Job-specific JSA reviewed at every shift change; daily toolbox talks

Isolation first

Valero's isolation and LOTO verified before any connection is made

Pressure control

Normal 150–300 psi; over-pressure checklist above 500 psi

Line of fire

Bleed to zero at every pig change; nobody in line with lids or hoses

Hose integrity

Tested & certified hard and flex pipe; whip checks on every connection

PPE & coil care

FRC and face shields; no galvanized metal on stainless

Engineered for safety Retractable cam-actuated stairs, composite work platform, LED lighting

08 CAPABILITIES

What sets USA DeBusk apart

Self-performed scope

Pigging, decoking and hydroblasting with our own equipment and crews: one contractor, no subs

Largest standardized fleet

6 identical TriMax units, 18 engines, with a spare pump on board even on a 2-pass heater

Widest pig selection

Engineered decoking pigs matched to each coil and each coke condition

Data-driven, shared

Durations built from recorded actuals; lessons learned reviewed with Valero after every job



Trusted across the Valero system



14 heaters cleaned at 5 Valero refineries

Texas City, TX
H-57 coker heater, 2023 and 2024

Port Arthur, TX
H-102A, H-102B, F-2000

Three Rivers, TX
100-H-1, 100-H-2, H-1102, H-1105, 440-H1

Meraux, LA
VU-H-001, VU-H-002

Wilmington, CA
20H200, 20H2000, 31H3000

Coker, vacuum, crude charge and reboiler service

Valero case study & lessons learned

Valero Port Arthur, TX: H-102A & H-102B

Vacuum heaters, March–April 2026

16	18,496'	434
passes, 2 heaters	coil footage	pigs run

Scope Both 8-pass vacuum heaters pigged in triple mode, with filtration and smart pig inspection of every pass.

Result Every pass signed clean and cleared by smart pig before rig-out. A mid-job TriMax swap rigged in within 2 hours, with no break in pigging.

Pass 6B: coke ring found by smart pig

Mixed-diameter vacuum pass, 3" to 8"

0.25"	24 h	188 psi
thick coke ring	dedicated pigging	flow-test reduction

Challenge Inspection found a circumferential coke ring at the inlet of a 5" tube, just before the step-up to 8".

Lesson learned On mixed-diameter passes, stage pigs for every diameter and plan time at each step-up before the inspection run.

10 WHY USA DEBUSK

Why USA DeBusk for Texas City



Proven at Valero

Your H-57 coker in 2023 and 2024, one of 14 Valero heaters we've cleaned

Close by

Fleet based in Deer Park, a short drive from Texas City

Resilient execution

Triple-pass TriMax: a spare pump on every 2-pass heater

Verified clean

4-point verification and joint sign-off on every pass

Safe by the numbers

EMR 0.48, ISNetworld A, zero fatalities in 3 years

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Thank you

Questions & discussion

Jason Harman | Commercial Manager | jharman@usadebusk.com | (713) 839-5961